

2026.3.25

数学作业纸

M·G
晨光文具

班级:

姓名:

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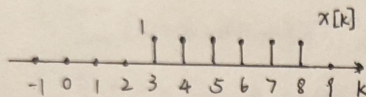
科目

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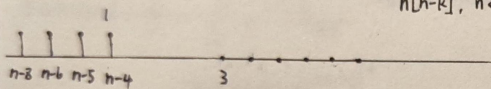
2.4 $y[n] = x[n] * h[n]$.

$$x[n] = \begin{cases} 1, & 3 \leq n \leq 8 \\ 0, & \text{其他} \end{cases}$$

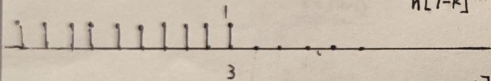
$$h[n] = \begin{cases} 1, & 4 \leq n \leq 15 \\ 0, & \text{其他} \end{cases}$$



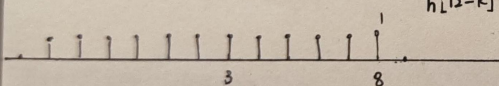
$$h[n-k], n < 7$$



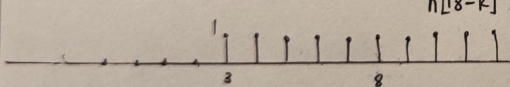
$$h[7-k]$$



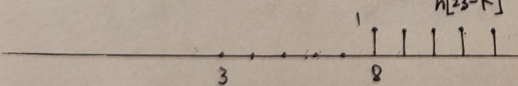
$$h[12-k]$$



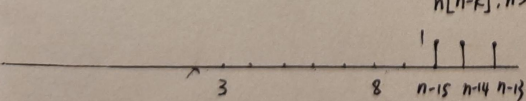
$$h[18-k]$$



$$h[23-k]$$



$$h[n-k], n > 23$$



$$\Rightarrow n < 7 \text{ 时, } y[n] = 0$$

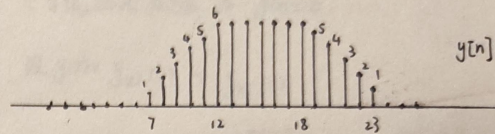
$$7 \leq n \leq 12 \text{ 时, } y[n] = n - 6$$

$$13 \leq n \leq 18 \text{ 时, } y[n] = 6$$

$$19 \leq n \leq 23 \text{ 时, } y[n] = 24 - n$$

$$n > 23 \text{ 时, } y[n] = 0$$

$$\Rightarrow y[n] = \begin{cases} n-6, & 7 \leq n \leq 12 \\ 6, & 13 \leq n \leq 18 \\ 24-n, & 19 \leq n \leq 23 \\ 0, & \text{其他} \end{cases}$$



2.11 $x(t) = u(t-3) - u(t-5)$ 且 $h(t) = e^{-3t} u(t)$

(a) 计算 $y(t) = x(t) * h(t)$.

$$y(t) = \int_{-\infty}^{\infty} x(\tau) h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} [u(\tau-3) - u(\tau-5)] e^{-3(t-\tau)} u(t-\tau) d\tau$$

当 $t < 3$ 时, $y(t) = 0$. 再分段

当 $3 \leq t \leq 5$ 时, $y(t) = \int_3^t e^{-3(t-\tau)} d\tau$

$$= e^{-3t} \int_3^t e^{3\tau} d\tau$$

$$= e^{-3t} \frac{1}{3} e^{3\tau} \Big|_3^t$$

$$= \frac{1}{3} e^{-3t} (e^{3t} - e^9)$$

$$= \frac{1 - e^{9-3t}}{3}$$

当 $t > 5$ 时, $y(t) = \int_3^5 e^{-3(t-\tau)} d\tau$

$$= e^{-3t} \int_3^5 e^{3\tau} d\tau$$

$$= e^{-3t} \frac{e^{15} - e^9}{3}$$

$$= \frac{e^{15-3t} - e^{9-3t}}{3}$$

$$\Rightarrow y(t) = \begin{cases} 0 & t < 3 \\ \frac{1 - e^{9-3t}}{3} & 3 \leq t \leq 5 \\ \frac{e^{15-3t} - e^{9-3t}}{3} & t \geq 5 \end{cases}$$

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(b) 计算 $g(t) = (dx(t)/dt) * h(t)$

$$\frac{d}{dt} x(t) = \delta(t-3) - \delta(t-5).$$

$$\Rightarrow g(t) = [\delta(t-3) - \delta(t-5)] * h(t)$$

$$= \int_{-\infty}^{\infty} [\delta(\tau-3) - \delta(\tau-5)] h(t-\tau) d\tau$$

$$= \int_{-\infty}^{\infty} [\delta(\tau-3) - \delta(\tau-5)] e^{-3(t-\tau)} u(t-\tau) d\tau.$$

当 $\tau=3$ 或 $\tau=5$, 且 $\tau \leq t$ 时有值.

且 $\tau=3$ 时 $h(t)$ 为 $e^{-3(t-3)}$, $\tau=5$ 时, 为 $-h(t)$.

$$\text{当 } t < 3 \text{ 时, } g(t) = \int_3^t e^{-3(t-\tau)} d\tau$$

$$= -\frac{1}{3} e^{-3(t-\tau)} \Big|_3^t$$

$$= -\frac{1}{3} e^{-3(t-t)} + \frac{1}{3} e^{-3(t-3)}$$

$$= \frac{e^{-3(t-3)}}{3}$$

$$\text{当 } t \geq 5 \text{ 时, } g(t) = \int_5^t e^{-3(t-\tau)} d\tau$$

$$\Rightarrow g(t) = \int_{\tau=3, \tau=5}^{\infty} e^{-3(t-\tau)} u(t-\tau) d\tau$$

$$= e^{-3(t-3)} u(t-3) - e^{-3(t-5)} u(t-5)$$

2.17

$$\frac{d}{dt} y(t) + 4y(t) = x(t) \quad \text{且满足初始松弛条件}$$

(a) 若 $x(t) = e^{(-1+3j)t} u(t)$, 求 $y(t)$.

$\therefore$ 系统初始松弛 $\Rightarrow y(0) = 0$.

$$\text{设 } y(t) = y_{zi}(t) + y_{zs}(t).$$

$$\Rightarrow y_{zi}(t) = y(t_0) e^{-a(t-t_0)} = 0 \cdot e^{-4(t-0)} = 0.$$

$$y_{zs}(t) = \int_{t_0}^t x(\tau) e^{-a(t-\tau)} d\tau$$

$$= \int_0^t e^{(-1+3j)\tau} e^{-4(t-\tau)} d\tau$$

$$= \int_0^t e^{(-1+3j)\tau} e^{-4t+4\tau} d\tau$$

$$= \int_0^t e^{-4t} \cdot e^{(3+3j)\tau} d\tau$$

$$= e^{-4t} \frac{1}{3+3j} e^{(3+3j)\tau} \Big|_0^t$$

$$= \frac{3-3j}{(3+3j)(3-3j)} e^{-4t} [e^{(3+3j)t} - 1]$$

$$= \frac{1-j}{6} [e^{(-1+3j)t} - e^{-4t}]$$

$\therefore$ 初始松弛, 即 $t < 0, y(t) = 0$.

$$\Rightarrow y(t) =$$

$$\Rightarrow y(t) = \frac{1-j}{6} [e^{(-1+3j)t} - e^{-4t}] u(t)$$

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(b) $\frac{d}{dt} y(t) + 4y(t) = x(t)$ 对 $\text{Re}\{x(t)\}$ 与 $\text{Re}\{y(t)\}$ 的关系也满足. 若 $x(t) = e^{-t} \cos(3t) u(t)$
求该线性时不变系统的输出 $y(t)$.

$$\therefore \frac{d}{dt} \text{Re}\{y(t)\} + 4 \text{Re}\{y(t)\} = \text{Re}\{x(t)\}.$$

$\Rightarrow$ 输入 $x_1(t) = \text{Re}\{x(t)\}$. 输出 $y_1(t) = \text{Re}\{y(t)\}$.

$$\begin{aligned} \therefore x_1(t) &= e^{-t} \cos(3t) u(t) = \text{Re}\{e^{-t} e^{j3t} u(t)\} \\ &= \text{Re}\left\{e^{-t} e^{j3t} u(t)\right\} = \text{Re}\left\{e^{(-1+j3)t} u(t)\right\} \\ &= \text{Re}\left\{e^{(1+j3)t} u(t)\right\}. \end{aligned}$$

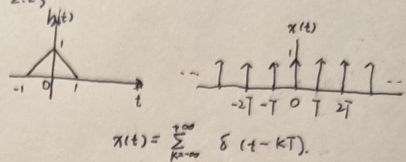
$$\Rightarrow y_1(t) = \text{Re}\left\{\frac{1-j}{6} [e^{(-1+j3)t} - e^{-4t}] u(t)\right\}$$

$$= \text{Re}\left\{\frac{1}{6} [e^{(-1+j3)t} - e^{-4t}] u(t) - j [e^{(-1+j3)t} - e^{-4t}] u(t)\right\}$$

$$= \text{Re}\left\{\frac{1-j}{6} e^{-t} [\cos(3t) + j \sin(3t)] - e^{-4t} u(t)\right\}$$

$$= \frac{1}{6} \{e^{-t} [\cos(3t) + \sin(3t)] - e^{-4t}\} u(t)$$

2.23



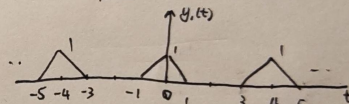
$$x(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT).$$

$$y(t) = x(t) * h(t).$$

(a) $T = 4$

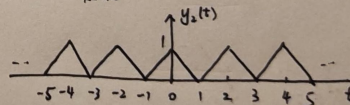
$$\begin{aligned} y(t) &= \left[\sum_{k=-\infty}^{+\infty} \delta(t - kT) \right] * h(t) \\ &= \sum_{k=-\infty}^{+\infty} h(t - kT) \end{aligned}$$

$$\Rightarrow y_1(t) = \sum_{k=-\infty}^{+\infty} h(t - 4k).$$



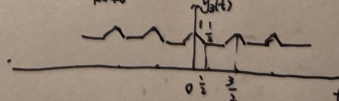
(b) $T = 2$

$$y_2(t) = \sum_{k=-\infty}^{+\infty} h(t - 2k)$$



(c) $T = \frac{3}{2}$

$$y_3(t) = \sum_{k=-\infty}^{+\infty} h(t - \frac{3}{2}k)$$



(d) $T = 1$

$$y_4(t) = \sum_{k=-\infty}^{+\infty} h(t - k)$$

